2025 6th International Conference on Machine Learning and Computer Application (ICMLCA) | 979-8-3503-9243-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICMLCA66850.2025.11336585

FRCA-Net: A Remote Sensing Image Fire Risk Classification Model Integrating Attention Mechanism and Auxiliary Detection

Aiguo Zhang
College of Computer and Information Engineering
Xiamen University of Technology
Xiamen, China
zhangaiguo@xmut.edu.cn

Lingyi Qi*
College of Computer and Information Engineering
Xiamen University of Technology
Xiamen, China
*Corresponding author: guyezhu@gmail.com

Zhenghang Yu
College of Computer and Information Engineering
Xiamen University of Technology
Xiamen, China

Jian Deng
College of Computer and Information Engineering
Xiamen University of Technology
Xiamen, China

***Abstract*—Wildfires, characterized by their sudden onset and destructive nature, pose severe threats to ecosystems and human safety. Efficient and accurate fire risk assessment is therefore of critical importance. Remote sensing technology, with its wide coverage and cost-effectiveness, provides valuable data for regional fire monitoring and risk evaluation. However, fire risk features in remote sensing images are often unstructured and low in contrast, making them difficult to extract. With the development of deep learning, its advantages in image recognition and classification have become increasingly evident in remote sensing applications. This study proposes an improved deep learning model, FRCA-Net, which integrates attention mechanisms and object detection modules to enhance the classification accuracy of seven fire risk categories in remote sensing imagery: Water, Non-burnable, Very Low, Low, Moderate, High, and Very High. The model adopts ResNet-50 as the backbone, incorporates the CBAM attention module to improve spatial focus, and utilizes the YOLOv8 object detection network to identify man-made structures for dynamically adjusting classification weights. Experiments on the public FireRisk dataset demonstrate that FRCA-Net achieves higher accuracy (0.6732), precision (0.5656), recall (0.5742), and F1 score (0.5558) than existing models such as ViT, DenseNet, and MAE. In particular, it shows more stable performance in medium- and low-risk areas and effectively reduces category confusion, leading to an overall accuracy improvement of approximately 4%. The proposed approach offers a practical and efficient solution for intelligent fire risk interpretation based on remote sensing imagery.**

Keywords: Convolutional block attention module(CBAM); convolutional neural networks(CNN); deep learning; fire risk assessment; remote sensing images.

I. INTRODUCTION

Forest fires are one of the major natural disasters affecting the ecological environment and socio-economic systems worldwide [1]. The occurrence of forest fires not only causes severe damage to forest resources [2], but also negatively impacts air quality, climate change, and human safety [3]. In recent years, climate change has intensified the frequency and severity of global fire activities [4]. For instance, the 2020 forest fire in Liangshan Prefecture, Sichuan Province, resulted in over 3,047 hectares of burned area, with 19 casualties and 3 injuries [5]. Similarly, a series of forest fires triggered by extreme heat and drought in Chongqing in 2022 posed serious threats to vegetation and damaged power and transportation infrastructure [6].

Fire risk zoning has proven effective in practical disaster prevention. Early identification of high-risk areas not only improves resource allocation efficiency but also significantly reduces the frequency of fire occurrences. Novo et al. used multi-source data and the Analytic Hierarchy Process (AHP) to delineate fire risk zones in Galicia, Spain, providing accurate management strategies for wildfire prevention along highways [7]. Chuvieco et al., in the Firemap project, integrated various fire-related factors to construct a spatial fire risk index, which was validated against historical fire records [8]. Chavan et al. classified fire risk levels in the Harnail watershed of the Himalayas and found that high-risk areas were effectively identified and prioritized, thus controlling the spread of wildfires [9]. These studies collectively demonstrate that fire risk classification not only holds academic value but also shows great potential for disaster mitigation in real-world scenarios. Therefore, accurate fire risk prediction and assessment are crucial for forest management, ecological protection, and emergency response [10][11]. Prior to fire outbreaks, risk level forecasting can provide scientific support for early warning and resource deployment, helping to minimize potential losses [12].

The rapid development of remote sensing technology has provided essential data support for fire risk classification. As a non-contact and large-scale observation method, remote sensing has been widely used in fire monitoring [13]. By analyzing the features of remote sensing imagery, fire risk in different regions can be effectively assessed [14]. Risk classification involves complex factors, including vegetation type [15], climate conditions, and topographic features [16].

Current methods for fire risk classification fall into two main categories. Binary classification distinguishes areas as “fire” or “non-fire” and is suitable for real-time fire monitoring and early warning due to its simplicity and fast response [17]. Multi-level classification, on the other hand, divides risk into several levels, enabling finer-grained evaluation and resource management [18]. However, there are still challenges in

This work was supported by the Natural Science Foundation of Fujian Province (Grant Nos. 2022J011237 and 2020J01262).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

applying remote sensing images for classification. For instance, land-cover heterogeneity may lead to spectral confusion between forests, grasslands, and buildings [19]. Additionally, limitations in spatial and temporal resolution may hinder the detection of rapidly changing risk factors, such as short-term weather variations [20]. Hence, achieving accurate classification of fire risk based on remote sensing imagery remains a major challenge in current research.

Many studies have applied traditional machine learning algorithms, such as Support Vector Machines (SVM), Random Forests, and Decision Trees, to classify remote sensing images [21]. For example, Quan X et al. used MODIS data along with various terrain, weather, and fuel variables to assess wildfire risk in China, employing Random Forest to analyze the correlations between variables and wildfires [22]. Kalantar B et al. adopted three machine learning models—MARS, SVM, and BRT—to predict forest fire susceptibility in the Chaloos Rood watershed of Iran [23]. While traditional machine learning algorithms can handle structured data, their ability to extract complex features from diverse remote sensing images is limited [24]. Since fire risk is associated with multiple interacting factors such as vegetation, climate, terrain, and spatial relationships, traditional methods often fall short in capturing these deeper characteristics.

With advancements in remote sensing, satellite imagery has become more detailed, and data volume has surged [25]. Deep learning and AI methods can leverage large-scale remote sensing datasets for training and learning. De Almeida Pereira G H et al. proposed a deep learning-based model using Landsat-8 imagery to develop a fire detection dataset with 150,000 image patches and demonstrated improved detection accuracy with CNN architectures [26]. However, this research focused more on fire detection than on fire risk assessment. Vuppari et al. employed Vision Transformer (ViT) to train on high-resolution images, achieving excellent wildfire recognition performance on the Wildfire dataset [27]. Huot et al. built predictive deep learning models using a decade of satellite data and fire history to identify high-risk fire zones [28]. Narvaez Luces et al. applied SPOT-6/7 imagery with a VQ-VAE-based unsupervised anomaly detection model to extract burned areas automatically [29]. Sun's study used multispectral remote sensing images from South America and trained a fully convolutional network (FCN) for pixel-wise fire detection, validating its cross-region adaptability [30]. These methods demonstrate the power of deep models in processing remote sensing images. However, some approaches still rely on non-image features, requiring additional domain knowledge and limiting model generalizability and transferability. Larsen et al. proposed an FCN-based model to detect smoke plumes in near-real-time remote sensing imagery, aiding fire exposure evaluation and public health communication [31]. Shen S et al. constructed a remote sensing image dataset [32] and explored the possibility of predicting fire risk solely from remote sensing imagery, such as proximity to human activities and forest species composition, indirectly reflecting fire risk levels.

Despite the progress in fire risk classification, accuracy remains a bottleneck. In particular, fine-grained classification tasks often suffer from performance limitations, making them unsuitable for practical deployment [33]. This study aims to address the multi-class classification of fire risk levels in remote sensing images using deep learning techniques. We propose a novel classification method that improves model structure and optimizes feature extraction, thereby overcoming limitations of traditional methods. Our approach fully exploits the complex features of remote sensing images by applying data augmentation to mitigate overfitting [34] and using deep neural networks to automatically extract multi-level information—including vegetation coverage, topographical variation, and spatial distribution—leading to more precise characterization of fire risk zones.

The remainder of this paper is organized as follows: Section 2 (Methodology) introduces the dataset construction, model architecture, and experimental settings; Section 3 (Experiments and Analysis) presents performance comparisons and ablation studies; Section 4 (Conclusion) summarizes the results and discusses limitations and future directions.

II. Method

A. Dataset

This study adopts the publicly available FireRisk remote sensing dataset to train the proposed deep learning model, FRCA-Net, which is specifically designed for fire risk classification in remote sensing imagery. The dataset was first introduced in [32], and the remote sensing images were collected from the high-resolution imagery provided by the National Agricultural Imagery Program (NAIP). The fire risk levels were labeled based on the Wildfire Hazard Potential (WHP) raster dataset. FireRisk contains seven fire risk categories: Water, Non-burnable, Very Low, Low, Moderate, High, and Very High. It comprises a total of 91,872 labeled images, each with a resolution of 320 × 320 pixels. Examples of the remote sensing images and their corresponding labels are shown in Fig. 1. The Non-burnable category mainly includes buildings and leveled artificial grasslands. The dataset has been widely used in the field of fire risk assessment.

The original FireRisk dataset consists of training and validation sets, with the validation set primarily used for performance evaluation during model training. Since there is no independent test set, the validation set may be used for both hyperparameter tuning and performance assessment, which could affect the objectivity of model generalization evaluation. To better assess the model's generalization ability, we re-partitioned the original validation set, allocating 70% as the new validation set and the remaining 30% as the test set. The distribution of fire risk categories in the dataset after partitioning is shown in Table I. The training set is used for learning model parameters, the validation set for tuning hyperparameters and monitoring overfitting, and the test set for final performance evaluation.

Similar processing is also found in other studies. For instance, in [35], Mahmood et al. re-partitioned subsets of the FireRisk dataset and merged the original seven fire risk categories into three broader classes—Non-burnable, High Risk, and Low Risk—and performed random sampling within each class to reduce dataset size and standardize the scale for multitask learning. In our experiments, to mitigate the risk of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

model overfitting caused by the scarcity of labeled remote sensing images in real-world applications [36][37], we applied various data augmentation techniques during training, including random flipping, scaling, cropping, and color variation, as illustrated in Fig. 2.

B. Base Model Architecture

The proposed FRCA-Net (Fire Risk Classification with CBAM and Auxiliary Detection) adopts ResNet-50 as the backbone classification network due to its outstanding performance in ImageNet image classification tasks [38]. The core of ResNet lies in its residual architecture, which employs skip connections to mitigate the vanishing gradient problem in deep neural networks. These connections allow the input to bypass multiple layers and be directly transmitted to later stages, enabling the effective training of very deep networks. This design improves training efficiency while significantly enhancing classification accuracy [38].

However, ResNet-50 also has certain limitations, particularly in capturing small-scale or subtle fire-related features [39]. While the network performs well in identifying high-risk regions with broad fire coverage, its capability to process fine-grained local variations requires further optimization. To address this, two enhancements are introduced to the ResNet-50 framework. First, a CBAM (Convolutional Block Attention Module) is incorporated after the middle and later residual blocks to strengthen the model's ability to selectively focus on deep features. Second, an auxiliary recognition module based on YOLOv8 is designed to specifically detect man-made structures within remote sensing images, structurally enhancing the model's discriminative power.

The YOLOv8 module is not an external detection tool but an integral component of the FRCA-Net architecture, participating directly in fire risk classification. It is responsible for localizing built-up areas—such as buildings and roads—and generating confidence scores C_{obj}. These scores are then fused

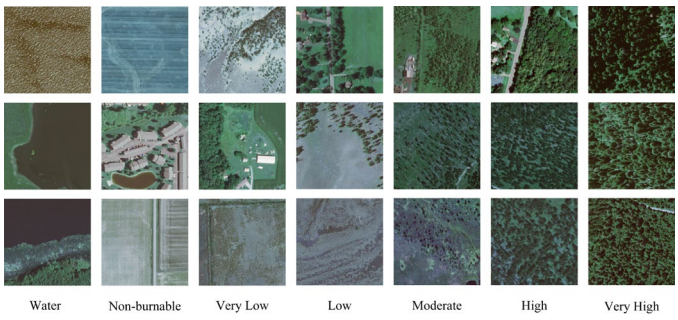


Figure 1. Example images and corresponding labels from the FireRisk dataset.



Figure 2. Example of data augmentation applied to a remote sensing image.

TABLE I. DISTRIBUTION OF FIRE RISK LABELS IN THE TRAINING, VALIDATION, AND TEST SETS

Label Name	Training Set	Validation Set	Test Set
Very Low	21757	5913	2535
Non-burnable	17959	3563	1528
Low	10705	1819	780
Moderate	8617	1240	532
High	6296	1126	483
Very High	3268	1006	432
Water	1729	408	176

with the risk probabilities output by the backbone network to enable dynamic adjustment of the final classification.

Specifically, analysis of remote sensing images revealed that certain low-risk categories (e.g., Low, Very Low, and Non-burnable) often contain a higher proportion of man-made structures such as buildings, roads, and bridges. A comprehensive analysis of the dataset shows that in the Very Low risk category, 21% of images include man-made structures, while in the Very High category, this proportion drops to just 0.4%. These features hold discriminative value in fire risk assessment [40] and provide supplementary information for classification. Therefore, we introduce an auxiliary recognition module to effectively leverage such structural cues and further refine the classification outcomes.

The auxiliary module uses the YOLOv8 object detection model [41] to analyze each image. When a high proportion of built-up structures is detected, the model automatically reduces the predicted fire risk weight; conversely, if few or no structures are detected, the risk weight remains unchanged or is increased. This mechanism enhances FRCA-Net's sensitivity to spatial scene structure, significantly improving both classification accuracy and model robustness.

Fig. 3 illustrates the overall architecture and processing pipeline of FRCA-Net, which consists of the DATA and MODEL components. Input remote sensing images first undergo data augmentation operations in the DATA module—including flipping, brightness/saturation variation, and cropping—to improve adaptability to complex scenarios. The augmented images are then processed by the MODEL module. The backbone ResNet-50 extracts multi-level features, while the CBAM module refines key representations within the deeper residual units. In parallel, the YOLOv8 module identifies structural features and outputs a confidence score C_{obj}, which is used to adjust the classification probability distribution. This fusion of structural information and category prediction enhances the overall effectiveness of fire risk identification.

C. Model Improvements

1) Network Structure Optimization:

The Convolutional Block Attention Module (CBAM) is a lightweight and efficient attention mechanism that integrates both channel and spatial attention [42]. It dynamically adjusts the weights of different channels and spatial positions during the feature extraction process, thereby enhancing the model's ability to emphasize informative regions while suppressing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

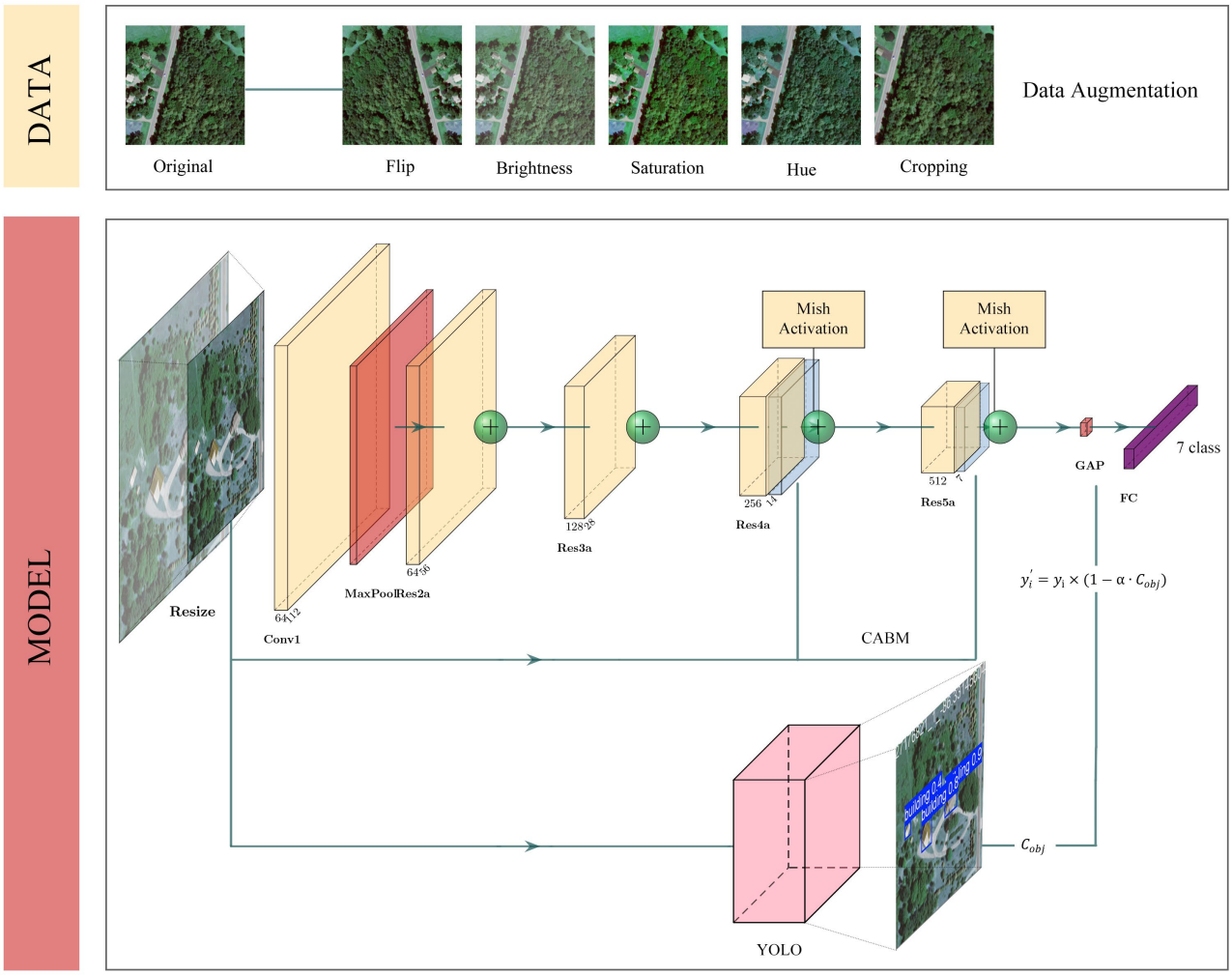


Figure 3. Overall architecture of the proposed FRCA-Net model, which integrates a ResNet-50 backbone with CBAM attention modules and an auxiliary YOLOv8 detection branch for identifying man-made structures.

irrelevant ones. CBAM consists of two sub-modules: the Channel Attention Module (CAM) and the Spatial Attention Module (SAM). The CAM applies global average pooling and max pooling across each input channel to compute their relative importance and adjust the weights accordingly. Meanwhile, the SAM aggregates feature responses via max and average pooling across the spatial dimension and generates attention maps that refine feature emphasis across different spatial locations.

In the fire risk classification task of this study, CBAM modules are integrated after the third (layer3) and fourth (layer4) residual blocks of ResNet, as shown in Fig. 4. These two layers correspond to the deeper stages of ResNet-50, where the spatial resolution of feature maps is lower but the number of channels is higher. They are primarily responsible for extracting high-level semantic information—layer3 capturing mid-scale features and layer4 focusing on more abstract, global representations.

By embedding CBAM in these two stages, the model's ability to regulate attention over deep semantic features is significantly strengthened. This enhancement enables more accurate identification of fire-prone regions in remote sensing imagery, particularly for small, localized, but semantically important high-risk features often obscured by complex backgrounds

2) Activation Function Optimization

Activation functions play a crucial role in neural networks, as they determine how input signals are transformed into nonlinear outputs. In the original ResNet architecture, the traditional Rectified Linear Unit (ReLU) is used. ReLU is simple and computationally efficient; however, it suffers from certain drawbacks, the most notable being the "dying ReLU" problem—where neurons become inactive and stop updating their weights when the input is negative, resulting in a loss of learning ability during training. Additionally, the ReLU function lacks smoothness, especially near zero, which can negatively impact model performance when dealing with complex datasets [43].

To address these issues, this study adopts the Mish activation function in place of ReLU [44]. Mish is a smooth, non-monotonic activation function defined as: as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

$$\text{Mish}(x)=x \bullet \tanh(\ln(1+e^{x})) \quad (1)$$

Compared with ReLU, Mish offers better smoothness, particularly near x=0, and produces more continuous outputs in the negative domain. It is unbounded above and asymptotically approaches a finite negative value below. The function curve of Mish is illustrated in Fig. 5. These properties allow the model to better capture complex patterns and features in challenging datasets. The smooth nature of Mish also promotes more stable gradient flow, reducing oscillations and instability during training [45].

Moreover, the use of Mish enhances the network's ability to capture small-scale features and improves the transmission of gradient information, thereby increasing the nonlinearity and expressive power of the model [44]. These advantages make Mish particularly suitable for complex classification tasks such as fire risk assessment in remote sensing imagery.

3) Auxiliary Detection Module

In fire risk classification tasks, information on man-made structures such as buildings and roads in remote sensing imagery is often correlated with specific fire risk categories [46]. Therefore, this study introduces the YOLOv8 model as an auxiliary detection module to identify man-made structures in remote sensing images, as shown in Fig. 6. When the auxiliary model detects such structures, the fire risk category probabilities are dynamically adjusted based on the detection results to lower the predicted risk level in the corresponding regions. The adjusted features are then passed into the main classification model.To incorporate the detection results of man-made structures from YOLOv8, we propose a confidence-based weight adjustment mechanism for refining the class probability outputs of the classification model.

In this module, the YOLOv8 model was initialized with publicly available pretrained weights trained on large-scale object detection datasets. To adapt YOLOv8 to the characteristics of the FireRisk dataset, we constructed a small subset of 100 images containing man-made structures (e.g., buildings and roads) sampled from FireRisk. These images were manually annotated with bounding boxes and used for light fine-tuning. This limited fine-tuning improves domain relevance while avoiding overfitting, given the auxiliary nature of this detection branch. Importantly, the YOLOv8 model does not participate in the training of FRCA-Net; it is only applied during inference, where its detection outputs are incorporated into the probability adjustment mechanism to refine fire risk predictions.

There are seven fire risk categories, and the prediction results are represented by a probability vector:

$$y=[y_1, y_2,..., y_7] \quad (2)$$

where yi denotes the predicted probability for fire risk category i. The YOLOv8 model returns a confidence score C_{obj}, which represents the confidence level of detected man-made regions in the image, with values ranging from 0 to 1. A higher C_{obj} indicates stronger evidence of man-made structures, while lower values imply weak or no detection.

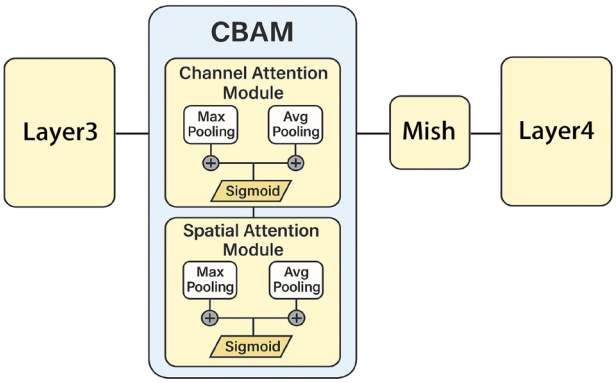


Figure 4. The architecture of the CBAM module, incorporated after layer3 and layer4 in FRCA-Net to enhance the network's ability to focus on informative semantic features for accurate fire risk classification.

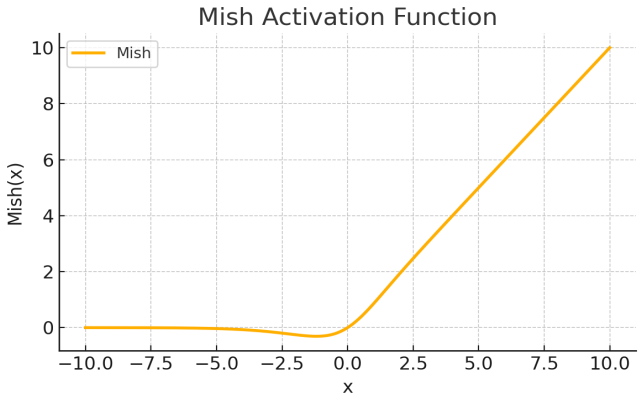


Figure 5. Visualization of the Mish activation function used in the proposed FRCA-Net model, illustrating its smooth, non-monotonic behavior that enhances gradient flow and improves feature extraction for complex fire risk classification tasks.



Figure 6. Visual examples of human-made structures detected in low-risk fire zones, demonstrating the contextual cues captured by the auxiliary detection module.

Each class probability y_i is adjusted using the following factor:

$$y'_i = y_i \bullet (1-\alpha \bullet C_{obj}) \quad (3)$$

where α is a hyperparameter controlling the degree of adjustment based on the detection confidence. The higher the confidence, the greater the downscaling applied to the original probability, aiming to reduce potential misclassification.

Unlike fixed-threshold methods, dynamic adjustment adapts to image content: few or no structures leave predictions unchanged; abundant high-confidence structures reduce predicted fire risk.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

III. Experiments and Analysis

To comprehensively evaluate the performance of the proposed remote sensing image classification model for fire risk assessment, we adopted several widely used classification metrics. These metrics help ensure that the model performs robustly across different types of fire risk regions.

During model training, a total of 91,872 remote sensing image samples were divided into training, validation, and test sets, comprising 70,331, 15,075, and 6,466 samples respectively. The training set was used to optimize the learnable parameters within the network, such as convolutional weights and biases. The validation set was employed for hyperparameter tuning, including settings such as the initial learning rate, batch size, and weight decay, as well as for monitoring overfitting during training. Finally, the test set was used to evaluate the model's generalization performance.

A. Experimental Setup

We used Visual Studio Code as the integrated development environment, with Python as the primary programming language and PyTorch as the deep learning framework. Detailed hardware and software configurations are listed in Table II. The backbone network adopted in this study was the improved ResNet-50, initialized with pretrained weights. ResNet is a deep convolutional neural network based on residual connections, and ResNet-50, as a representative implementation, employs a bottleneck residual structure to expand the network depth to 50 layers. Compared to shallower variants, ResNet-50 provides enhanced feature extraction capability [38], making it well suited for large-scale image classification tasks and providing a strong foundation for the subsequent modules.

All remote sensing images were resized to 224 × 224 pixels and augmented using random flipping and color jittering to improve the model's robustness. We trained the model using the Adam optimizer [47] with a weight decay of 1×10-4 , and set the initial learning rate to 0.01—slightly higher than the default value for Adam—to accelerate early-stage convergence when using pretrained weights. To further stabilize the learning process and reduce gradient oscillation, we employed a cosine annealing learning rate scheduler [48], which decayed the learning rate in a cosine pattern every 10 epochs. The batch size was set to 128. During training, the model was validated after each epoch on the validation set.The effectiveness of the proposed improvements was evaluated through rigorous testing using multiple metrics, including classification accuracy, precision, recall, F1 score, and confusion matrix analysis. To comprehensively assess classification performance, we compared FRCA-Net with several baseline models using these metrics, aiming to reveal the strengths and weaknesses of different network architectures in the context of fire risk classification from remote sensing imagery.

TABLE II. Experimental Environment and Configuration.

Category	Specification
Operating System	Ubuntu 22.04.3 LTS
CPU	Intel Core i3-12100KF
GPU	NVIDIA GeForce RTX 3090
System Memory	64GiB
Framework	PyTorch 0.19.1 with CUDA 12.1

B. Models

To verify the effectiveness of the proposed FRCA-Net model in fire risk classification tasks, we selected four representative image classification models as baselines: ResNet-50, DenseNet, Vision Transformer (ViT), and Masked Autoencoder (MAE). These models are widely used in remote sensing image classification and natural image understanding, representing the mainstream deep learning architectures from different paradigms.

ResNet-50 [38] is a classical convolutional neural network (CNN) architecture that mitigates the vanishing gradient problem in deep networks by introducing residual connections. It has become a standard benchmark model for image classification tasks. DenseNet [39] adopts a densely connected structure where each layer is connected to all preceding layers, promoting feature reuse and reducing the number of parameters. It has shown excellent performance in tasks with limited training data, such as remote sensing image classification.ViT (Vision Transformer) [49] is a pure attention-based architecture that divides an image into patches and models their relationships through self-attention mechanisms. It overcomes the local receptive field limitation of traditional CNNs and has become a state-of-the-art (SOTA) model in image understanding.MAE (Masked Autoencoder) [50] is a self-supervised visual representation learning method that learns generic features by randomly masking patches in input images and reconstructing the missing content. As a representative of the pretraining-and-finetuning paradigm, MAE has achieved remarkable results across various downstream tasks.

For fairness, we applied the same input resolution, pretrained weights, finetuning epochs, and data augmentation as used in FRCA-Net for all baseline models.

TABLE III. Classification Performance On The Test Set: Comparison Of Common Models Vs. The Proposed Model.

Model	Accuracy	Precision	Recall	F1 Score
ResNet-50	0.6320	0.5494	0.5351	0.5336
ViT	0.6331	0.5460	0.5375	0.5223
DenseNet	0.6275	0.5324	0.5336	0.5317
MAE	0.6502	0.5531	0.5492	0.5465
FRCA-Net	0.6732	0.5656	0.5742	0.5558

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

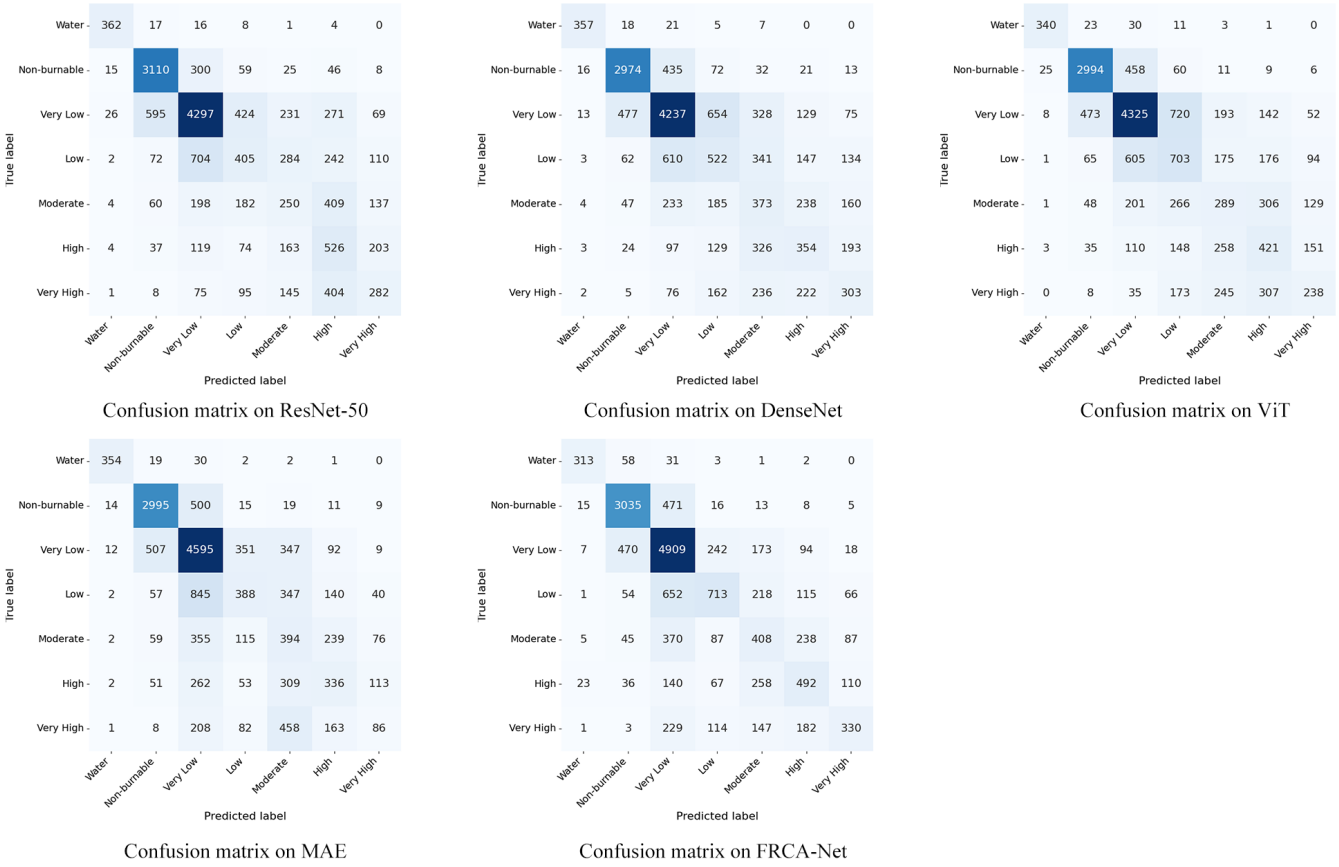


Figure 7. Confusion matrices of different models on the fire risk classification task, illustrating category-wise prediction performance and misclassification patterns. FRCA-Net shows improved overall accuracy and robustness, particularly on low-risk classes such as “Non-burnable” and “Low”.

C. *Model Performance Comparison*

To evaluate the performance of FRCA-Net in fire risk classification, we conducted comparative experiments with the four baseline models mentioned above. The results are summarized in Table III. In terms of classification accuracy, FRCA-Net achieved the highest score of 0.6732, outperforming MAE by 0.0230 and ResNet-50 by 0.0412, indicating a significant improvement. For precision, FRCA-Net reached 0.5656, slightly higher than MAE’s 0.5531 and approximately 0.0332 higher than DenseNet. Regarding recall, FRCA-Net achieved 0.5742, clearly surpassing the other models, with an improvement of about 0.0250 over MAE and 0.0391 over ResNet-50. As for the F1 score, FRCA-Net obtained 0.5558, which is 0.0093 higher than MAE’s 0.5465, and remains the highest among all models. Overall, FRCA-Net outperformed the baselines across all four metrics, with particularly notable gains in accuracy and recall, demonstrating its strength in improving both overall recognition and the identification of potential risk areas. These results indicate that fire risk classification performance can be significantly enhanced through architectural optimization without introducing additional data.

To further analyze the classification performance of each model, we visualized their confusion matrices, as shown in Fig. 7. A comparison of the matrices reveals notable differences in category-wise prediction capability among models. The diagonal elements represent correctly classified samples, while the off-diagonal elements indicate misclassifications, helping to pinpoint specific strengths and weaknesses. FRCA-Net performs well on low-risk categories such as “Non-burnable” and “Low,” with fewer misclassifications and overall higher precision across other categories as well. In contrast, models like ResNet-50 and ViT exhibit relatively poor performance on low-risk categories, leading to more frequent class confusion and lower overall recall and precision.

D. *Effect Of Attention Mechanism*

To guide the model’s focus toward salient regions of the image rather than distributing attention uniformly, we incorporated an attention mechanism. To intuitively demonstrate the contribution of the CBAM module to classification performance, we generated visualizations of the attention weight distributions. To verify whether attention enhancement improves the discrimination of critical categories, we further analyzed the classification performance on high-risk categories, as summarized in Table IV. After integrating CBAM, the model achieved improvements in overall accuracy, recall, and F1 score. In particular, the recall for high-risk categories increased from 0.4972 to 0.5389, indicating that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

attention mechanism helps the model better identify key fire-prone areas. As shown in Fig. 8, CBAM assigns higher attention to high-risk areas (forests, dry grasslands) and lower weights to background elements (urban buildings, roads), highlighting key regions and suppressing irrelevant information, effectively improving classification accuracy [51].

E. Impact Of Training Strategies

In this section, we systematically evaluate the impact of three core training optimization strategies—data augmentation, learning rate scheduling, and optimizer selection—on the performance of the fire risk classification model. These experiments demonstrate how these strategies collectively contribute to improving classification accuracy, training stability, and convergence speed. Specifically, three sub-experiments are designed to quantify the effectiveness of each strategy using evaluation metrics including Accuracy, Recall, and F1 Score. The purpose of each sub-experiment is as follows: (1) Data augmentation: This experiment investigates how different augmentation techniques enhance the model's ability to resist overfitting and improve generalization; (2)Learning rate scheduling: We compare fixed learning rates, step decay, and cosine annealing schedulers to evaluate their influence on training efficiency and convergence behavior;(3) Optimizer selection: This part analyzes how different optimizers affect model training stability and final classification performance.

1) Network Structure Optimization:

Data augmentation is an effective approach to enhance model generalization, especially for remote sensing images, where sample complexity and diversity are high. In this study, we evaluated the effects of several common augmentation techniques—such as random flipping, flipping, color jittering, cropping, and their combinations—on fire risk classification performance. To this end, we conducted controlled experiments with and without data augmentation and recorded performance metrics including Accuracy, Recall, and F1 Score. The results are summarized in Table V, which shows that all augmentation strategies led to performance improvements to varying degrees.

TABLE IV. IMPACT OF THE CBAM MODULE ON HIGH-RISK CATEGORY RECALL AND OVERALL CLASSIFICATION PERFORMANCE.

Model	Accuracy	Recall	F1 Score	High-Risk Recall
Original	0.6320	0.5351	0.5336	0.4972
With CBAM	0.6517	0.5506	0.5493	0.5389

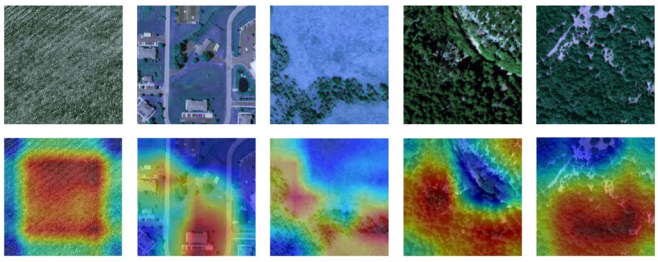


Figure 8. Visualization of the attention mechanism. Top: original image; bottom: attention map overlay

Compared to the "no augmentation" baseline, using only flipping improved accuracy by 0.24 percentage points (from 0.6173 to 0.6197) and increased the F1 score by 1.73 percentage points (from 0.5054 to 0.5227). Color jittering alone yielded even more notable gains, raising accuracy to 0.6278 and F1 score to 0.5362.

Among the combination strategies, "flipping + color jittering" achieved the best overall performance, with accuracy reaching 0.6320—1.47 percentage points higher than the baseline—and F1 score increasing by 2.82 percentage points. These results indicate that flipping helps the model adapt to spatial transformations, while color jittering simulates different imaging and climatic conditions. The combination of both enables the model to better cope with variations in image appearance, thereby improving classification accuracy and reducing overfitting risk.

2) Learning Rate Scheduling:

Learning rate scheduling dynamically adjusts the learning rate during training to accelerate convergence and improve final model performance. In this study, we evaluated three common learning rate scheduling strategies: fixed learning rate, step decay, and cosine annealing. The classification results under each strategy are summarized in Table VI.

The results show clear differences in classification performance and convergence behavior among the three methods. The fixed learning rate strategy achieved an Accuracy of 0.6182 and an F1 score of 0.5207, delivering stable but limited improvement, with relatively slow convergence. In contrast, the step decay scheduler reduced the learning rate by a factor of 0.1 every 5 epochs, mitigating instability caused by an excessively large learning rate in early training. This strategy improved the Accuracy to 0.6226 and increased the F1 score to 0.5372—an improvement of 1.65 percentage points over the fixed strategy—while also achieving faster convergence.

The cosine annealing scheduler further optimized the training process by allowing larger learning rates in the early phase to speed up convergence and gradually reducing the rate in later stages to suppress oscillations. This method achieved an Accuracy of 0.6306 and an F1 score of 0.5267. Although the F1 score was slightly lower than that of the step decay scheduler, cosine annealing demonstrated higher overall training stability and efficiency. Training loss curves are shown in Fig. 9.

TABLE V. IMPACT OF DIFFERENT DATA AUGMENTATION STRATEGIES ON MODEL PERFORMANCE.

Augmentation Strategy	Accuracy	Recall	F1 Score
None	0.6173	0.5058	0.5054
Flip	0.6197	0.5193	0.5227
Color Jitter	0.6278	0.5393	0.5362
Crop	0.6176	0.5153	0.5154
Flip + Crop	0.6226	0.5413	0.5341
Flip + Color Jitter	0.6320	0.5351	0.5336
Flip + Crop + Color Jitter	0.6204	0.5287	0.5307

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

TABLE VI. EFFECT OF DIFFERENT LEARNING RATE SCHEDULING STRATEGIES ON MODEL PERFORMANCE.

Strategy	Accuracy	Recall	F1 Score
Fixed Learning Rate	0.6182	0.5286	0.5207
Step Learning Rate	0.6226	0.5377	0.5372
Cosine Learning Rate	0.6306	0.5319	0.5267



Figure 9. Training loss curves under different learning rate schedules (Fixed, Step, and Cosine). Cosine annealing leads to faster and smoother convergence.

3) Optimizer Selection:

The choice of optimizer plays a crucial role in determining training efficiency and final model performance. To assess the impact of different optimizers on the fire risk classification task using remote sensing imagery, we compared three commonly used optimization algorithms: Stochastic Gradient Descent (SGD), Adadelta, and Adam. The experimental results are summarized in Table VII.

As shown in the results, SGD provides stable performance, achieving an Accuracy of 0.6141 and an F1 Score of 0.5138. While its overall performance is moderate, it requires more training epochs to reach convergence due to slower early-stage learning. Adadelta demonstrates faster convergence but slightly lower overall performance, with an Accuracy of 0.6001 and an F1 Score of 0.5074. Although Adadelta benefits from adaptive learning, its ability to model high-dimensional and complex features appears limited in this task.

In contrast, the Adam optimizer achieved the best performance across all evaluation metrics. It reached an Accuracy of 0.6320, a Recall of 0.5351, and an F1 Score of 0.5336, significantly outperforming the other two optimizers. Adam's superior results are attributed to its combination of momentum and adaptive learning rate mechanisms, which allow it to dynamically adjust update steps across varying gradient scales, enabling more effective learning of diverse risk regions in remote sensing images.

Fig. 10 presents the training loss curves for the three optimizers. As illustrated, Adam exhibits faster loss reduction in the early stages and a smoother convergence trajectory, further confirming its suitability and advantage for this classification task.

To summarize the findings from this section, the combination of random flipping and color jittering, cosine annealing learning rate scheduling, and the Adam optimizer yields the best overall performance in terms of accuracy, recall, and F1 score. This configuration also demonstrates improved training stability and faster convergence. These results highlight the importance of appropriate data augmentation, dynamic learning rate adjustment, and optimizer selection in enhancing model generalization and training efficiency.

F. *Ablation Study*

To further validate the contributions of individual components in the proposed model architecture, we conducted ablation experiments by progressively removing or replacing key modules (e.g., CBAM and the auxiliary recognition module) and observing their impact on overall performance. These experiments help assess the importance of each module and their effect on classification accuracy. The results are summarized in Table VIII.

When the CBAM module was removed, the accuracy dropped from 0.6517 to 0.6320, indicating that CBAM effectively enhances the model's focus on critical regions in the image and improves its ability to distinguish high-risk areas. Further, when the auxiliary recognition module was excluded, the accuracy slightly decreased to 0.6486. Although the drop in accuracy was modest, both recall and F1 score experienced more significant declines, suggesting that this module plays an important role in improving the model's ability to identify low-risk categories (such as regions with man-made structures).

This finding is further supported by the confusion matrices shown in Fig. 11. Taking the "Very Low" category as an example, the number of correctly predicted instances dropped from 4909 (with the recognition module) to 4635 (without it). Similar accuracy declines were observed in other categories, demonstrating the positive contribution of the auxiliary module in recognizing non-fire-prone regions. By integrating building detection results from YOLOv8 into the classification process, we introduced a confidence-based dynamic weighting mechanism. This novel approach significantly improves classification precision. The recognition module helps the model better distinguish between man-made structures and natural landscapes, lowering the predicted fire risk levels in such areas and avoiding overestimation.

TABLE VII. EFFECT OF DIFFERENT OPTIMIZERS ON MODEL PERFORMANCE.

Optimizer	Accuracy	Recall	F1 Score
SGD	0.6141	0.5099	0.5138
Adadelta	0.6001	0.5069	0.5074
Adam	0.6320	0.5351	0.5336

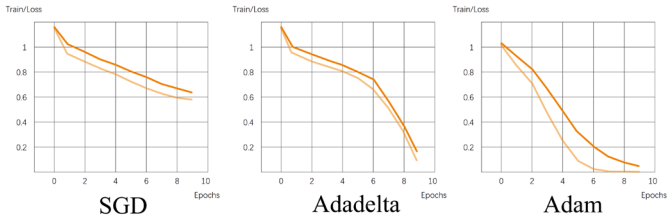
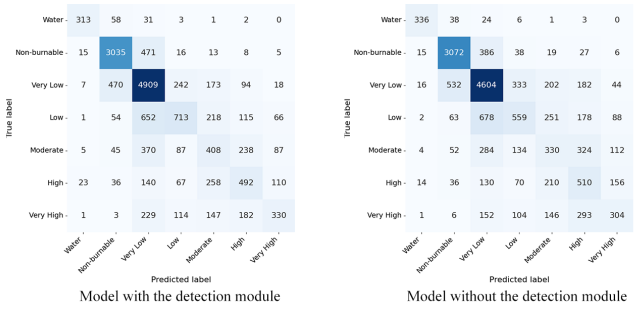


Figure 10. Training loss curves for different optimizers. Adam converges faster and more stably to a lower loss compared to SGD and Adadelta.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII. ABLATION STUDY: EFFECT OF CBAM AND DETECTION MODULES ON MODEL PERFORMANCE.

Model Configuration	Accuracy	Recall	F1 Score
Baseline Model	0.6320	0.5351	0.5336
+ CBAM	0.6517	0.5506	0.5493
+ Detection	0.6486	0.5537	0.5423
+ CBAM and Detection	0.6732	0.5742	0.5558



Model with the detection module

Model without the detection module

Figure 11. Confusion matrices of the full model (with auxiliary detection) and the variant without it. The auxiliary module improves classification accuracy, especially for low-risk classes, by increasing true positives along the diagonal.

Overall, the CBAM and auxiliary recognition modules play complementary roles—CBAM improves attention to high-risk areas, while the auxiliary module enhances discrimination in low-risk regions. Their combined use significantly boosts the overall classification performance

IV. CONCLUSION

This study proposes FRCA-Net, a novel model that integrates the CBAM attention mechanism and a YOLOv8-based auxiliary detection module to improve fire risk classification in remote sensing images without relying on additional external data. The model adopts ResNet-50 as the backbone network, ensuring stable training and effective feature extraction in deep architectures. The CBAM module enhances the model's focus on critical regions while suppressing background noise, thus improving the identification of diverse fire risk features. The YOLOv8 module contributes by incorporating information on man-made structures, enhancing the model's understanding of complex geographic scenes and reducing misclassifications in non-critical areas. These enhancements collectively improve the model's adaptability to diverse image conditions and regions, thereby strengthening its generalization on heterogeneous remote sensing datasets.

Experimental results demonstrate that incorporating CBAM increases the model's accuracy to 0.6517, with Recall and F1 Score both improving by approximately 0.015, especially enhancing the recognition of high-risk areas. The further integration of the YOLOv8 module boosts the accuracy to 0.6732, indicating better overall performance and generalization in multi-class classification tasks. Despite these improvements, FRCA-Net still presents opportunities for optimization in terms of computational efficiency and deployment cost. Future work may focus on lightweight strategies to enable real-world applications in resource-constrained environments.

Two main directions are proposed for future research. First, an auxiliary model targeting tree species recognition could be introduced to enhance fire risk classification in complex forest environments. Although the current auxiliary module effectively detects human-made structures, its performance in natural vegetation scenes remains limited. Second, to address the high computational cost of FRCA-Net, lightweight designs should be explored to accelerate inference and improve resource efficiency. Future efforts may involve developing more efficient architectures or adopting model compression techniques such as pruning and quantization, aiming to reduce computational load while maintaining accuracy for practical deployment.

ACKNOWLEDGMENT

This work was supported by the Natural Science Foundation of Fujian Province (Grant Nos. 2022J011237 and 2020J01262).

REFERENCES

[1] A. Saleh, M. A. Zulkifley, H. H. Harun, F. Gaudreault, I. Davison, and M. Spraggon, "Forest fire surveillance systems: A review of deep learning methods," Heliyon, vol. 10, no. 1, Jan. 2024, Art. no. e23127, doi: 10.1016/j.heliyon.2023.e23127.

[2] M. A. Moritz, E. Batllori, R. A. Bradstock, A. M. Gill, J. Handmer, and P. F. Hessburg, et al., "Learning to coexist with wildfire," Nature, vol. 515, no. 7525, pp. 58–66, Nov. 2014, doi: 10.1038/nature13946.

[3] H. Bargali, A. Pandey, D. Bhatt, R. C. Sundriyal, and V. P. Uniyal, "Forest fire management, funding dynamics, and research in the burning frontier: A comprehensive review," Trees, Forests and People, vol. 16, Apr. 2024, Art. no. 100526, doi: 10.1016/j.tfp.2024.100526.

[4] I. Prapas, A. Ahuja, S. Kondylatos, I. Karasante, E. Panagiotou, L. Alonso, et al., "Deep Learning for Global Wildfire Forecasting," arXiv preprint, arXiv:2211.00534, Nov. 2022.

[5] People's Government of Sichuan Province, "Investigation report on the '3·30' forest fire incident in Xichang, Liangshan Prefecture," Sichuan Provincial Government Portal, Dec. 21, 2020. [Online].

[6] Y. Z. Zhou, Q. Li, Y. G. Wu, "Research on urban fire rescue in forest fire suppression," Forest Fire Prevention, vol. 41, no. 2, pp. 36–39, 2023. (in Chinese)

[7] A. Novo, N. Fariñas-Álvarez, J. Martínez-Sánchez, H. González-Jorge, J. M. Fernández-Alonso, H. Lorenzo, et al., "Mapping forest fire risk — A case study in Galicia (Spain)," Remote Sens., vol. 12, no. 22, p. 3705, Nov. 2020, doi: 10.3390/rs12223705.

[8] E. Chuvieco, I. Aguado, M. Yebra, H. Nieto, J. Salas, M. P. Martin, et al., "Development of a framework for fire risk assessment using remote sensing and geographic information system technologies," Ecol. Model., vol. 221, no. 1, pp. 46-58, Jan. 2010, doi: 10.1016/j.ecolmodel.2008.11.017.

[9] M. E. Chavan, K. K. Das, and R. S. Suryawanshi, "Forest fire risk zonation using remote sensing and GIS in Huynial watershed, Tehri Garhwal district, UA," J. Indian Soc. Remote Sens., Jan. 2012.

[10] F. Di Giuseppe, C. Vitolo, G. Barnard, G. Libertá, P. Maciel, J. San-Miguel-Ayanz, et al., "Global seasonal prediction of fire danger," Scientific Data, vol. 11, no. 1, Art. no. 128, Jan. 2024, doi: 10.1038/s41597-024-02948-3.

[11] K. Lee, W. J. D. van Leeuwen, J. K. Gillan, and D. A. Falk, "Examining the impacts of pre-fire forest conditions on burn severity using multiple remote sensing platforms," Remote Sensing, vol. 16, no. 10, May 2024, Art. no. 1803, doi: 10.3390/rs16101803.

[12] M. Naderpour, H. M. Rizeei, and F. Ramezani, "Forest fire risk prediction: A spatial deep neural network-based framework," Remote Sensing, vol. 13, no. 13, Jul. 2021, Art. no. 2513, doi: 10.3390/rs13132513.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

[13] D. M. Szpakowski and J. L. R. Jensen, “A review of the applications of remote sensing in fire ecology,” Remote Sensing, vol. 11, no. 22, Nov. 2019, Art. no. 2638, doi: 10.3390/rs11222638.

[14] C. Magro, O. C. Gonçalves, L. Nunes, and J. M. C. Pereira, “Remote sensing of volatile organic compounds release during prescribed fires in pine forests using open-path Fourier transform infra-red spectroscopy,” Int. J. Wildland Fire, vol. 33, no. 4, 2024, doi: 10.1071/WF23019.

[15] W. Kang and Y. Kim, “Extraction of fire risk factors near power transmission facilities using high-resolution satellite images,” in Proc. IGARSS 2024–IEEE Int. Geosci. Remote Sens. Symp., Athens, Greece, Jul. 2024, pp. 3831–3835.

[16] C. Güngöroğlu, İ. İsmailoğlu, B. Kapukaya, and M. Y. Şahin, “Comparison between post-fire analysis and pre-fire risk assessment according to various geospatial data,” Sustainability, vol. 16, no. 4, Feb. 2024, Art. no. 1569, doi: 10.3390/su16041569.

[17] Z. Xu, J. Li, and L. Xu, “Wildfire risk prediction: A review,” arXiv preprint, arXiv:2405.01607, 2024. [Online]. Available: https://arxiv.org/abs/2405.01607

[18] İ. Ersoy, E. Ünsal, and Ö. Gürsoy, “A multi-criteria forest fire danger assessment system on GIS using literature-based model and analytical hierarchy process model for Mediterranean Coast of Manavgat, Türkiye,” Sustainability, vol. 17, no. 5, p. 1971, Mar. 2025, doi: 10.3390/su17051971.

[19] W. Liu, S. Gopal, and C. E. Woodcock, “Uncertainty and confidence in land cover classification using a hybrid classifier approach,” Photogramm. Eng. Remote Sens., vol. 70, no. 8, pp. 963–971, 2004.

[20] Y. Zhai, Z. Qu, and L. Hao, “Land cover classification using integrated spectral, temporal, and spatial features derived from remotely sensed images,” Remote Sens., vol. 10, no. 3, p. 383, 2018, doi: 10.3390/rs10030383.

[21] A. E. Maxwell, T. A. Warner, and F. Fang, “Implementation of machine-learning classification in remote sensing: An applied review,” Int. J. Remote Sens., vol. 39, no. 9, pp. 2784–2817, May 2018, doi: 10.1080/01431161.2018.1433343.

[22] X. Quan, Q. Xie, B. He, and X. Wang, “Corrigendum to: Integrating remotely sensed fuel variables into wildfire danger assessment for China,” Int. J. Wildland Fire, vol. 30, no. 10, p. 822, Oct. 2021, doi: 10.1071/WFv30n10_CO.

[23] B. Kalantar, N. Ueda, M. O. Idrees, and B. K. Pradhan, “Forest fire susceptibility prediction based on machine learning models with resampling algorithms on remote sensing data,” Remote Sensing, vol. 12, no. 22, Nov. 2020, Art. no. 3682, doi: 10.3390/rs12223682.

[24] Rewhel E. M., Li J., Hamed A. A., Keshk H. M., Mahmoud A. S., Sayed S. A., et al., “Deep learning methods used in remote sensing images: A review,” J. Environ. Earth Sci., vol. 5, no. 1, pp. 33–64, Apr. 2023, doi: 10.30564/jees.v5i1.5232.

[25] W. Jiang, Y. Sun, L. Lei, and G. Kuang, “Change detection of multisource remote sensing images: A review,” Int. J. Digital Earth, vol. 17, no. 1, pp. 1–24, Jan. 2024, doi: 10.1080/17538947.2024.2398051.

[26] G. H. De Almeida Pereira, A. M. Fusioka, B. T. Nassu, and D. Menotti, “Active fire detection in Landsat-8 imagery: A large-scale dataset and a deep-learning study,” ISPRS J. Photogramm. Remote Sens., vol. 178, pp. 171–186, Jan. 2021, doi: 10.1016/j.isprsjprs.2021.06.010.

[27] R. Vuppari, C. Park, and A. Srivastava, “Forest Fire Detection in Remote Sensing Images Using Vision Transformers,” arXiv preprint, arXiv:2505.17395, 2025. [Online]. Available: https://arxiv.org/abs/2505.17395

[28] F. Huot, A. Capderou, and F. Landry, “Wildfire Hazard Prediction with Earth Observation Data and Deep Learning,” arXiv preprint, arXiv:2010.07445, 2020. [Online]. Available: https://arxiv.org/abs/2010.07445

[29] M. Narvaez Luces, E. Pezzulo, and L. Oneto, “Unsupervised Burned Area Detection Using SPOT-6/7 Satellite Imagery and Vector Quantized Variational Autoencoders,” arXiv preprint, arXiv:2308.13367, 2023. [Online]. Available: https://arxiv.org/abs/2308.13367

[30] L. Sun, “Pixel-Level Forest Fire Detection Using FCN in Multispectral Satellite Images in South America,” arXiv preprint, arXiv:2201.09671, 2022. [Online]. Available: https://arxiv.org/abs/2201.09671

[31] Larsen A., Hanigan I., Reich B. J., Qin Y., Cope M., Morgan G., et al., “A deep learning approach to identify smoke plumes in satellite imagery in near-real time for health risk communication,” J. Expo. Sci. Environ. Epidemiol., vol. 31, no. 1, pp. 170–176, Jul. 2021, doi: 10.1038/s41370-020-0246-y.

[32] S. Shen, S. Seneviratne, X. Wanyan, and Y. Zhang, “Firerisk: A remote sensing dataset for fire risk assessment with benchmarks using supervised and self-supervised learning,” in Proc. 2023 Int. Conf. Digital Image Comput.: Tech. Appl. (DICTA), Dec. 2023, pp. 189–196, doi: 10.1109/DICTA56561.2023.10312345.

[33] Huot F., Hu R. L., Ihme M., Wang Q., Burge J., Lu T., et al., “Deep learning models for predicting wildfires from historical remote-sensing data,” arXiv preprint, arXiv:2010.07445, Oct. 2020. [Online]. Available: https://arxiv.org/abs/2010.07445

[34] R. Archana and P. S. E. Jeevaraj, “Deep learning models for digital image processing: A review,” Artif. Intell. Rev., vol. 57, no. 1, pp. 11–35, Jan. 2024, doi: 10.1007/s10462-023-10456-7.

[35] S. Mahmood, J. Yao, and S. W. Kim, “Multi-LoRA Meets Vision: Merging Multiple Adapters to Create a Multi-Task Model,” arXiv preprint, arXiv:2403.13149, 2024. [Online]. Available: https://arxiv.org/abs/2403.13149

[36] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “ImageNet classification with deep convolutional neural networks,” in Advances in Neural Information Processing Systems, vol. 25, 2012, pp. 1097–1105.

[37] C. Shorten and T. M. Khoshgoftaar, “A survey on image data augmentation for deep learning,” J. Big Data, vol. 6, no. 1, pp. 1–48, Dec. 2019, doi: 10.1186/s40537-019-0197-0.

[38] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., Jun. 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.

[39] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), Honolulu, HI, USA, Jul. 2017, pp. 4700–4708.

[40] S. Gui, S. Song, R. Qin, and Y. Zhang, “Remote sensing object detection in the deep learning era—a review,” Remote Sensing, vol. 16, no. 2, Jan. 2024, Art. no. 327, doi: 10.3390/rs16020327.

[41] J. Redmon, “You only look once: Unified, real-time object detection,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), Las Vegas, NV, USA, Jun. 2016, pp. 779–788.

[42] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, “CBAM: Convolutional block attention module,” in Proc. Eur. Conf. Comput. Vis. (ECCV), Munich, Germany, Sep. 2018, pp. 3–19.

[43] Y. Zhang, K. M. Daud, K. Moorthy, and M. A. S. M. Yusof, “Filtering approaches and Mish activation function applied on handwritten Chinese character recognition,” Adv. Distrib. Comput. Artif. Intell. J., vol. 13, 2024, Art. no. e31218.

[44] D. Misra, “Mish: A self regularized non-monotonic activation function,” arXiv preprint, arXiv:1908.08681, 2019. [Online]. Available: https://arxiv.org/abs/1908.08681

[45] Chakravarthy S. R. S., Bharanidharan N., Vinothini C., Kumar V. V., Mahesh T. R., Guluwadi S., *et al.*, “Adaptive Mish activation and ranger optimizer-based SEA-ResNet50 model with explainable AI for multiclass classification of COVID-19 chest X-ray images,” *BMC Med. Imaging*, vol. 24, no. 1, Art. no. 206, Aug. 2024, doi: 10.1186/s12880-024-01394-2.

[46] K. V. Thakare and K. M. Tajne, “Urban fire risk hot spot assessment and green spaces: A geospatial analysis of fire risk mitigation in Nagpur,” Cureus J. Eng., vol. 1, no. 1, Aug. 2024, Art. no. 69, doi: 10.7759/s44388-024-00069-6.

[47] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” arXiv preprint, arXiv:1412.6980, 2014. [Online]. Available: https://arxiv.org/abs/1412.6980

[48] I. Loshchilov and F. Hutter, “SGDR: Stochastic gradient descent with warm restarts,” *arXiv preprint*, arXiv:1608.03983, 2016. [Online]. Available: https://arxiv.org/abs/1608.03983

[49] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16×16 words:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.

Transformers for image recognition at scale," arXiv preprint arXiv:2010.11929, 2020. [Online]. Available: https://arxiv.org/abs/2010.11929

[50] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick, "Masked autoencoders are scalable vision learners," arXiv preprint arXiv:2111.06377, 2021. [Online]. Available: https://arxiv.org/abs/2111.06377

[51] Y. Xu, J. Li, L. Zhang, H. Liu, and F. Zhang, "CNTCB-YOLOv7: An effective forest fire detection model based on ConvNeXtV2 and CBAM," *Fire*, vol. 7, no. 2, Feb. 2024, Art. no. 54, doi: 10.3390/fire7020054.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:36 UTC from IEEE Xplore. Restrictions apply.